

# Waqas Sultani

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| CONTACT INFORMATION | Email: waqas.sultani@itu.edu.pk, Mobile: +92-3365109108,<br>Personal Website      MedAI Lab Website      LinkedIn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| RESEARCH SUMMARY    | My research focuses on computer vision and multimodal AI, with an emphasis on foundation models for digital hematology and medical image analysis. I have also made contributions to visual surveillance, video understanding, and cross-view geolocalization. My work has been recognized through the Google Research Scholar Award and the Facebook Computer Vision for Global Challenges Research Award, and has appeared in CVPR, TPAMI, IJCV, MICCAI, Journal of Medical Image Analysis, AAAI, WACV, and ICRA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ACADEMIC POSITION   | <div><div><b>Associate Professor</b></div><div>2024-Present</div><ul style="list-style-type: none"><li>Department of Artificial Intelligence, Information Technology University, Lahore</li><li>Head of MedAI research group at Intelligent Machine Lab, ITU.</li></ul></div> <div><div><b>Assistant Professor</b></div><div>2018-2024</div><ul style="list-style-type: none"><li>Computer Science Department, Information Technology University, Lahore</li><li>Head of MedAI research group at Intelligent Machine Lab, ITU.</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| EDUCATION           | <div><div><b>Ph.D. Computer Science (Computer Vision),</b></div><div>2011-2017</div><ul style="list-style-type: none"><li>Center for Research in Computer Vision, University of Central Florida, USA</li><li>Advisor: Dr. Mubarak Shah, Trustee Chair Professor</li><li>Dissertation: Weakly Labeled Action Recognition and Detection</li></ul></div> <div><div><b>MS, Electrical Engineering and Computer Science,</b></div><div>2008-2010</div><ul style="list-style-type: none"><li>Seoul National University, South Korea</li><li>Perceptual and Intelligence lab</li><li>Advisor: Dr. Jin Young Choi</li></ul></div> <div><div><b>BSc. Electrical Engineering,</b></div><div>2003-2006</div><ul style="list-style-type: none"><li>University of Engineering and Technology, Taxila, Pakistan</li><li>Major: Electronics</li><li>Advisor: Dr. Muhammad Amin</li></ul></div>                                                                                                                                                                                                                                                                  |
| POSITIONS           | <div><div><b>Assistant Professor</b></div><div>2018- 2024</div><ul style="list-style-type: none"><li>Information Technology University, Lahore</li></ul></div> <div><div><b>Lead Computer Vision Scientist</b></div><div>2023-2024</div><ul style="list-style-type: none"><li>Obvio, USA</li></ul></div> <div><div><b>Principal Machine Learning Engineer</b></div><div>2022-2023</div><ul style="list-style-type: none"><li>Hazen.ai, Saudi Arabia: Took a one-year leave from ITU to gain industrial research experience</li></ul></div> <div><div><b>Graduate Research Assistant</b></div><div>2011-2017</div><ul style="list-style-type: none"><li>University of Central Florida, USA</li></ul></div> <div><div><b>Summer Research-Intern</b></div><div>2014</div><ul style="list-style-type: none"><li>Xerox, New York, USA</li></ul></div> <div><div><b>Lecturer</b></div><div>2010-2011</div><ul style="list-style-type: none"><li>COMSATS University, Islamabad</li></ul></div> <div><div><b>System Design Engineer</b></div><div>2007-2008</div><ul style="list-style-type: none"><li>Advanced Engineering Organization</li></ul></div> |

## AWARDS, GRANTS

- **Google**, Research Scholar Award, “A Large-Scale Blood Cancer Dataset of Multiple Microscopes for Low-Cost Blood Cancer Detection”, Role: Lead PI
- **Facebook AI**, Computer Vision for Global Challenges research award, “Low-cost deep learning solution to real-time detection of malaria”, Role: Lead PI. • **HEC, Startup Research Grant** Program, “Criminal activities detection in CCTV video using computer vision techniques”, Role: Lead PI
- **HEC, Crime Investigation and Prevention Lab**, Role: Co-PI project funded by HEC
- **MICCAI Open Data Micro-Grant** Role: Lead PI
- **Among Top 2% of Researchers Worldwide** Included in the list of top scientists by researchers (top 2% of Researchers Worldwide) from Stanford University led by Dr. John PA Ioannidis in terms of productivity and impact in the years 2026.
- Recipient of **NSF-USA** Innovation-corps training
- **Travel Grants** for CVPR and ICPR
- **MS scholarship** by Higher Education Commission, Pakistan, 2010

## PATENTS

- Robert P. Loce, **Waqas Sultani**, Beilei Xu, Hao Wu, “System and Method for Seat Occupancy Detection from Ceiling Mounted Camera using Robust Adaptive Threshold Criteria”, **US Patent: US9378421 B2**.
- Robert P. Loce, **Waqas Sultani**, Hao Wu, Beilei Xu, Thomas F. Wade, Mary Ann Sprague, Patricia Swenton-Wall, Megan Clar, Eric Harte, “System and Method for Detecting Settle Down Time using Computer Vision Techniques”, **US Patent: US9384396 B2**.

## PROFESSIONAL SERVICE

- Area Chair in CVPR 2022, 2023, 2024, 2025, 2026, 2027
- Area Chair in ICLR, 2027
- Area Chair in Neurips 2024, 2025
- Area Chair in ECCV 2026
- Area Chair in WACV 2024, 2025, 2026
- Area Chair in ACM Multimedia Conference, 2021, 2020
- Reviewer of ECCV, IEEE TIP, CVIU, ICCV
- Program Committee of WACV, HADCV workshop, 2021, DL-HAU2020, ICPR, 2020, and Frontiers of Information Technology (FIT)

## SELECTED PUBLICATIONS

- A. Abu Lehyeh, Xiaohan Zhang, A. Arrabi, **Waqas Sultani**, Chen Chen, Safwan Wshah, “GeoFlow: Real-Time Fine-Grained Cross-View Geolocalization via Iterative Flow Prediction”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2026.
- A. Rehman, I. Rasool, A. Imran, Mohsen Ali, **Waqas Sultani**, “Uni-Hema: Unified Model for Digital Hematopathology”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2026.
- Y. Chen, Xiaohan Zhang, A. Arrabi, **Waqas Sultani**, Chen Chen, Safwan Wshah, “Turning Generators into Retrievers: Unlocking MLLMs for Natural Language-Guided Geo-Localization”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, **CVPRW**, 2026.
- Hajra Malik, Hafiza Tooba Aftab, Abdul Rehman, Mohsen Ali, **Waqas Sultani**, “Multilingual Hematology Visual Question Answering Dataset”, *MICCAI Open Dataset workshop Workshops*, **MICCAIW**, 2026.
- A. Arrabi, Xiaohan Zhang, **Waqas Sultani**, Chen Chen, Safwan Wshah, “Cross-view Meets Diffusion: Aerial Image Synthesis with Geometry and Text Guidance”, *IEEE/CVF Winter Conference on Applications of Computer Vision*, **WACV**, 2025.

- Abdul Rehman, Talha Meraj, Aiman Mahmood Minhas, Ayisha Imran, Mohsen Ali, **Waqas Sultani**, Mubarak Shah, “Leveraging Sparse Annotations for Leukemia Diagnosis on the Large Leukemia Dataset”, *Medical Image Analysis, MIA*, 2025, Impact Factor: 14
- Xiaohan Zhang, Xingyu Li, **Waqas Sultani**, Chen Chen, Safwan Wshah “GeoDTR+: Toward generic cross-view geolocalization via geometric disentanglement ”, *IEEE Transaction on Pattern Analysis and Machine Intelligence, IEEE TPAMI*, 2024, Impact Factor: 18.5
- Muhammad Atif Butt, Hassan Ali, Adnan Qayyum, **Waqas Sultani**, Ala Al-Fuqaha, Junaid Qadir “R2S100K: Road-Region Segmentation Dataset For Semi-Supervised Autonomous Driving in the Wild”, *International Journal of Computer Vision, IJCV*, 2024, Impact Factor: 11.6
- Abdul Rehman, Talha Meraj, Aiman Mahmood Minhas, Ayisha Imran, Mohsen Ali, **Waqas Sultani**, “A Large-scale Multi Domain Leukemia Dataset for the White Blood Cells Detection with Morphological Attributes for Explainability”, *27th International conference on Medical Image Computing and Computer-assisted Intervention MICCAI*, 2024
- Sumayya Inayat, Nimra Dilawar, **Waqas Sultani**, Mohsen Ali, “Few-Shot Domain Adaptive Object Detection for Microscopic Images”, *27th International conference on Medical Image Computing and Computer-assisted Intervention MICCAI*, 2024
- Tushar Sangam, Ishan Rajendrakumar Dave, **Waqas Sultani**, Mubarak Shah “Transvisdrone: Spatio-temporal transformer for vision-based drone-to-drone detection in aerial videos ”, *IEEE International Conference on Robotics and Automation, ICRA*, 2023
- Xiaohan Zhang, Xingyu Li, **Waqas Sultani**, Yi Zhou, Safwan Wshah “Cross-view Geo-localization via Learning Disentangled Geometric Layout Correspondence ”, *AAAI Conference on Artificial Intelligence, AAAI*, 2023
- Xiaohan Zhang, **Waqas Sultani**, Safwan Wshah “Cross-View Image Sequence Geo-localization ”, *IEEE Winter Conference on Applications of Computer Vision, WACV*, 2023
- Daniel Wilson, Xiaohan Zhang, **Waqas Sultani**, Safwan Wshah “Image and Object Geo-Localization”, *International Journal of Computer Vision, IJCV*, 2023, Impact Factor: 19.5, HJRS: W
- Ifrah Siddiqui, Suleman Mazhar, Naufil Hassan, **Waqas Sultani** “Fine-Grained Road Quality Monitoring Using Deep Learning”, *IEEE Transactions on Intelligent Transportation Systems, ITS*, 2023, Impact Factor: 8.5 HJRS: W
- Javed Iqbal, Aliza Masood, **Waqas Sultani**, Mohsen Ali “Leveraging topology for domain adaptive road segmentation in satellite and aerial imagery”, *Journal of Photogrammetry and Remote Sensing, ISPRS*, 2023, Impact Factor: 12.7, HJRS: W
- Muaaz Zakria, Hamza Rawal, **Waqas Sultani**, Mohsen Ali “Cross-region building counting in satellite imagery using counting consistency”, *Journal of Neural Computing and Applications, NCA*, 2023, Impact Factor: 6, HJRS: W
- **Waqas Sultani**, Wajahat Nawaz, Syed Javed, Muhammad Sohail Danish, Asma Saadia, Mohsen Ali “Towards Low-Cost and Efficient Malaria Detection ”, *IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2022
- M Fasi Ur Rehman, Izza Aftab, **Waqas Sultani**, Mohsen Ali “Mapping Temporary Slums From Satellite Imagery Using a Semi-Supervised Approach”, *IEEE Geoscience and Remote Sensing Letters, IEEE GRSL*, 2022, Impact Factor: 4.8, HJRS: W

- Adnan Qayyum, **Waqas Sultani**, Fahad Shamshad, Rashid Tufail, Junaid Qadir “Single-shot retinal image enhancement using untrained and pretrained neural networks priors integrated with analytical image priors”, *Journal Computers in Biology and Medicine*, **CBM**, 2022, Impact Factor: 6.698, HJRS: W
- Muhammad Zaid, Shafaqat Ali, Mohsen Ali, Sarfaraz Hussein, Asma Saadia, **Waqas Sultani** “Identifying out of distribution samples for skin cancer and malaria images”, *Journal of Biomedical Signal Processing and Control*, **BSPC**, 2022, Impact Factor: 5, HJRS: W
- Bilal Yousaf, Muhammad Usama, **Waqas Sultani**, Arif Mahmood, Junaid Qadir “Fake visual content detection using two-stream convolutional neural networks”, *Journal of Neural Computing and Applications*, **NCA**, 2022, Impact Factor: 5.1, HJRS: W
- Qazi Ammar Arshad, Mohsen Ali, Saeed-ul Hassan, Chen Chen, Ayisha Imran, Ghulam Rasul, **Waqas Sultani** “A dataset and benchmark for malaria life-cycle classification in thin blood smear images”, *Journal of Neural Computing and Applications*, **NCA**, 2022, Impact Factor: 5.1, HJRS: W
- Waseem Ashraf, **Waqas Sultani**, Mubarak Shah. “Dogfight: Detecting Drones from Drones Videos”, *IEEE Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2021
- Nadeem Yousaf, Sarfaraz Hussein, **Waqas Sultani** “Estimation of BMI from facial images using semantic segmentation based region-aware pooling”, *Journal Computers in Biology and Medicine*, **CBM**, 2021, Impact Factor: 4.6, HJRS: W
- **Waqas Sultani**, Mubarak Shah “Human Action Recognition in Drone Videos using a Few Aerial Training Examples”, *Journal of Computer Vision and Image Understanding*, **CVIU**, 2021, Impact Factor: 3.9, HJRS: W
- Adnan Qayyum, **Waqas Sultani**, Fahad Shamshad, Junaid Qadir, and Rashid Tufail, “Single-Shot Retinal Image Enhancement Using Deep Image Priors”, *International Conference on Medical Image Computing and Computer-Assisted Intervention*, **MICCAI**, 2020
- Usman Ali, **Waqas Sultani**, Mohsen Ali, “Destruction detection from sky: Weakly supervised approach for destruction detection in satellite imagery”, *Journal of Photogrammetry and Remote Sensing*, **ISPRS**, 2020, 162, pp.115-124, Impact Factor: 7.3
- Anza Shakeel, **Waqas Sultani**, Mohsen Ali, “Deep built-structure counting in satellite imagery using attention based re-weighting”, *Journal of Photogrammetry and Remote Sensing*, **ISPRS**, 2019, 151, pp.313-321, Impact Factor: 6.9
- **Waqas Sultani**, Chen Chen, Mubarak Shah, “Real-world anomaly detection in surveillance videos”, *IEEE Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2018
- **Waqas Sultani**, Mubarak Shah, “Automatic Action Annotation in Weakly Labeled Videos”, *Journal of Computer Vision and Image Understanding*, **CVIU**, 2017, Impact Factor: 2.8, HJRS: W
- **Waqas Sultani**, Dong Zhang, Mubarak Shah, “Unsupervised Action Proposal Ranking through Proposal Recombination”, *Journal of Computer Vision and Image Understanding*, **CVIU**, 2017, Impact Factor: 2.8, HJRS: W
- **Waqas Sultani**, Soroush Mokhtari and Hae-Bum Yun, “Automatic Pavement Object Detection using Superpixel Segmentation Combined with Conditional Random Field”, *IEEE Transactions on Intelligent Transportation Systems*, **IEEE Trans ITS**, 2017, Impact Factor: 4.0, HJRS: W
- **Waqas Sultani**, Mubarak Shah, “What if we do not have multiple videos of the same action? - Video Action Localization Using Web Images”, *IEEE Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2016
- **Waqas Sultani**, Imran Saleemi, “Human Action Recognition across Datasets by Foreground Focused Histogram Decomposition”, *IEEE Conference on Computer Vision and Pattern Recognition*, **CVPR**, 2014

- **Waqas Sultani**, Jin Young Choi, “Abnormal Traffic Detection using Intelligent Driver Model”, *IEEE International Conference on Pattern Recognition, ICPR, 2010*
- **Waqas Sultani**, Jin Young Choi, “Abnormality Detection in Traffic Scenes”, *Summer Conference of Electronics Engineering Society of Korea, 2010*

#### INVITED TALKS

- ‘Large Scale Multi-Microscope Datasets and their Challenges’, at Stanford MedAI Group, USA
- ‘Toward Low-Cost microscopy’, at Birmingham City University, UK
- National Guest Speaker at International Integrated Health Science and Research Conference.
- ‘Unified Models for Digital Hematology’, at International Symposium on Data Technologies
- ‘Low-cost Malaria and Leukemia Diagnostics’, at Conference on Public Health and public policies
- ‘CVPR 2023 workshop: A Comprehensive Tour and Recent Advancements toward Real-world Visual Geo-Localization
- ‘Real-world anomaly detection in Surveillance videos’ at Lahore University of Management Sciences, Pakistan
- ‘Recognizing human actions on novel datasets’ at Xerox Research Center, USA
- ‘Weakly Labeled Action Detection’ at Florida Institute of Technology, USA
- ‘Weakly Labeled Human Action Recognition’ at Punjab University College of Information Technology, Pakistan

#### TEACHING

Teaching is one of the most rewarding aspects of my academic career. It not only allows me to share knowledge but also continually strengthens my own understanding by challenging me to revisit fundamental concepts and explore the finer details of the subject. To cultivate independent and logical thinking among students, I employ non-traditional learning techniques such as project-based learning, cooperative learning, and project competitions. Actively involving students in presenting their projects before the class serves to enhance their presentation and communication skills, instill confidence, clarify concepts, and necessitate a dedicated investment of time in the course. These projects can potentially evolve into undergraduate and master’s theses, enabling students to contribute significantly and produce research papers.

At Information Technology University (ITU), I have taught a wide range of undergraduate and graduate courses, including Medical Image Computing, one-to-one Computer Vision, Artificial Neural Networks and Deep Learning, Advanced Algorithm Analysis, and Theory of Automata. I continuously refine my teaching methods to align with emerging technologies, industry needs, and advances in artificial intelligence. My goal is to create an intellectually stimulating classroom that encourages curiosity, analytical thinking, creativity, and lifelong learning while equipping students with both theoretical foundations and practical problem-solving skills. My commitment to excellence in teaching is reflected in the consistently high student evaluation scores I have received throughout my tenure at ITU.

#### RESEARCH INTERESTS

My research interests span computer vision, machine learning, and multimodal AI. My early research focused on sports video understanding and intelligent surveillance, while my recent work has concentrated on medical image analysis and digital pathology. I have developed learning algorithms for challenging real-world settings, including few-shot learning, domain adaptation, weakly supervised learning, and low-resource medical imaging. My current research aims to develop multimodal foundation models and efficient small vision-language models for digital hematology, enabling a unified framework for disease diagnosis, cell understanding, report generation, and clinical decision support. I am also interested in extending foundation models to other vision problems, including geospatial understanding and cross-view geolocalization.

#### RESEARCH LEADERSHIP

Since joining Information Technology University (ITU), I have focused on building a sustainable and internationally competitive research ecosystem in Artificial Intelligence for Healthcare. I established the Medical AI Lab (Med-AI) to bring together undergraduate and graduate students working on computer vision, medical image analysis, and AI-driven healthcare solutions. To strengthen the educational foundation supporting this research vision, I introduced new undergraduate and graduate courses in Medical Image Computing, integrating contemporary topics in deep learning, computer vision, and biomedical AI. Under my supervision, undergraduate, master’s, and PhD students have published papers in premier international venues, including CVPR, MICCAI, Medical Image Analysis (MedIA) and other leading conferences and journals. To date, I have supervised 25 MS theses, 3 PhD students, and numerous undergraduate research projects.

My research program has been supported by the Google Research Scholar Award and the Facebook Research Award, enabling the acquisition of computing infrastructure and funding for graduate students. To broaden research opportunities, I launched a Summer Research Internship Program that trains talented students from universities across Pakistan in AI research and scientific writing. These initiatives have helped establish a collaborative research culture that emphasizes innovation, high-impact publications, and the development of the next generation of AI researchers.

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| BOOK CHAPTERS     | <ul style="list-style-type: none"><li>• <b>Waqas Sultani</b>, Qazi Ammar Ahmad, Chen Chen “Action Recognition in Real-World Videos”, <i>Computer Vision: A Reference Guide, Springer Link, 2020</i></li><li>• Sijie Zhu, Chen Chen, <b>Waqas Sultani</b> “Video Anomaly Detection for Smart Surveillance”, <i>Computer Vision: A Reference Guide, Springer Link, 2020</i></li></ul>                                    |
| STUDENTS          | <ul style="list-style-type: none"><li>• 01 PhD student graduated (Co-Supervisor)</li><li>• 02 PhD students currently under-supervision</li><li>• 20+ MS students graduated</li><li>• 30+ undergraduate students graduated</li></ul>                                                                                                                                                                                    |
| ACADEMIC SERVICES | <ul style="list-style-type: none"><li>• Graduate Evaluation Committee member of several PhD and MS students.</li><li>• Member of the Doctoral Program Review Committee</li><li>• Member of the Curriculum Design Program Review Committee</li><li>• Member of graduate/undergraduate Admissions committee.</li><li>• Part of the activity is to provide constructive feedback to new-hired teaching fellows.</li></ul> |